

HARSHDIP SAHA

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Education

Netaji Subhas University of Technology - 8.78 GPA

Bachelor of Technology in Computer Science and Engineering

Aug. 2023 – Aug. 2027

Delhi, India

Kendriya Vidyalaya No.2, Delhi Cantt - 95.6%

Class XII

April 2022 – April 2023

Delhi, India

Kendriya Vidyalaya No.2, Delhi Cantt - 98.4%

Class X

April 2020 – April 2021

Delhi, India

Relevant Coursework

- Computer Vision
- Natural Language Processing
- Game theory
- Artificial Intelligence
- DBMS
- Machine Learning

Experience

Research Intern - IIT Madras

Jan 2026 – Present

Scientific Computing

Remote

- Developing **pySdf**, a Python library for **Signed Distance Functions (SDFs)** enabling simulation-ready geometric representations from images, CAD models, and analytical inputs.
- Selected for oral presentation at INCAM 2026, at IIT Kanpur, then to be published in **Scopus** indexed journal.
- Built a scalable geometry processing engine with **CSG operations** and **STL-to-SDF conversion**, integrated with **AMReX MultiFab** for high-performance simulations using **GPU-accelerated bindings**.
- Applied **Chan-Vese segmentation** and 3D morphometric analysis to enable biomedical imaging workflows, extracting simulation-ready structures from volumetric medical data.

AI Engineer Intern - Optum

Jun 2026 – Aug 2026

Artificial Intelligence & Machine Learning

Bangalore, India

- I will be working on large real healthcare datasets and LLMs.

Projects

AToM-Net: Neuro-Symbolic LLM Negotiation Agent | *LLMs, NLP, Bayesian Inference, Game Theory* **Feb 2026**

- Designed a **neuro-symbolic NLP architecture** for multi-turn negotiation, combining **LLM-based dialogue understanding** with a **Bayesian belief tracking module** to model opponent intent.
- Introduced an **Economic Consistency Validator** to detect contradictions between natural language ("cheap talk") and actions ("costly signals"), improving reasoning robustness.
- Implemented a **Reflexion-based memory module** enabling inter-dialogue learning without weight updates, improving strategic adaptation across conversations.
- Achieved **23.30 vs 12.53 average score** against baseline LLMs on the CaSiNo negotiation benchmark, demonstrating superior language-grounded decision making.
- First architectural work to present results on the NegotiationTOM dataset, achieving SOTA performance with an overall score of **16.29**. ([Paper](#)).

RECAP-Net: Classification of Active Progression | *PyTorch, MONAI, GANs, Swin UNETR, 3D CNNs* **Sept 2025**

- Proposed an end-to-end pipeline for the **BraTS-PRO 2025 Challenge**, performing longitudinal glioblastoma response classification under the **RANO criteria**, integrating **Swin UNETR** segmentation with an ensemble of 3D ResNet, DenseNet, and EfficientNet classifiers.
- Applied **z-score normalization**, multimodal fusion of baseline/follow-up MRIs, and a **3D spectrally normalized GAN** for class balancing, achieving **3D-SSIM: 0.92 ± 0.03** and **KL divergence: 0.037 ± 0.004** .
- Accepted for **oral presentation at MICCAI 2025**, a top-tier (A*) conference in medical image computing.
- Achieved **94.0% balanced accuracy**, **94.6% F1-score**, and **96.0% AUROC**, outperforming individual backbones ([Paper](#)).

Brand-Aware Diffusion for Automated Marketing Poster Generation | *NLP, CLIP, Diffusion Models*

Jan 2026

- Developed a **brand-aware generative system** that combines **diffusion models (Stable Diffusion XL)** with **LLM-based text generation** for automated marketing poster creation.
- Designed a pipeline integrating **CLIP-based semantic understanding** and **context-aware NLP generation** for headlines, subheadlines, and CTAs based on user prompts.
- Implemented **context detection (festival, product launch, sales)** using LLM inference, enabling adaptive multimodal generation aligned with brand identity.
- Built a FastAPI backend and Adobe Express add-on enabling **interactive text editing and layer-wise control** over generated outputs.
- This project **won the Adobe** enterprise hackathon.([Paper](#)).

AI Loan Assistant – WhatsApp & Website Chatbot | *NLP, XGBoost, Twilio API, Firebase, OCR* **Dec 2025**

- Developed an AI-powered loan origination platform enabling users to apply for loans through both Website Chat and WhatsApp using a centralized multi-agent orchestration system.
- Built automated KYC verification, OCR-based document parsing, and credit risk assessment pipelines using **XGBoost** for real-time loan approval or rejection decisions.
- Integrated **Twilio WhatsApp API**, multilingual support using DeepTranslator, and sanction letter PDF generation with ReportLab for automated delivery via WhatsApp or website download ([Demo Video](#)).

Accurate Precise Timely | *Machine Learning, XGBoost, LSTM, ARIMA, Greedy scheduling* **Mar 2025**

- Developed an AI-powered system that schedules doctor appointments based on patient condition severity and predicted consultation time, while aligning with doctor availability and shift timings.
- Built an admin panel for real-time hospital resource monitoring and implemented LSTM and ARIMA models to predict next-day needs, refined daily using staff updates ([Demo Video](#)).
- Selected among the **top 30** teams to pitch offline at the **AI4Humanity Summit** (with the Israel Embassy) in a hackathon of 150+ teams([Certificate](#)).

Technical Skills

Languages: Python, C/C++, [MATLAB](#), SQL, JavaScript, HTML/CSS

Frameworks and Tools: Tensorflow, OpenCV, PyTorch, Pygame, Scikitlearn, pytesseract (OCR), Tkinter, Streamlit, SHAP (xAI), Git, Azure, GCP, spaCy, nltk, LangChain, Jupyter, statsmodels, Visual Studio Code

Databases and Services: Azure Blob Storage, MySQL, PostgreSQL, Hugging Face, [Kaggle](#)

Core Competencies: [Computer Vision](#) , ML, Data Analysis, Natural Language Processing, Cloud Computing, Time Series Analysis, Deep Learning, [AI](#)

Key Achievements and Extracurriculars

- Secured **World Rank 3** in the **BraTS Lighthouse 2025 – Tumor Progression Challenge**, with our paper accepted for **Oral Presentation** at the **MICCAI 2025 Conference, South Korea**.
- Currently working under the **NextGen AI Lab, NSUT**, contributing to advanced research including glioblastoma response classification using multimodal MRI data, NLP and deep learning architectures.
- Top open-source contributor to [BrainGlobe](#) ecosystem; merged **15+ pull requests** across repositories and a leading contributor to [brainglobe-registration](#), improving neuroscience image registration pipelines.
- Active ML member at Google Developer Students Club (GDSC) and core member at Hackethix Society.
- Secured All India 14th rank in BRAINDEAD Unstop competition by making two projects: one on EDA and ML, and another on an LLMs hybrid summarizer in Feb 2025.([GitHub Link](#)) ([Certificate Link](#))
- Solved **1000+** problems across various coding platforms such as [LeetCode](#), [GeeksforGeeks](#), and others. Achieved **Pupil** rank on [Codeforces](#).
- Ranked in the top 0.43% out of 1.1M students in JEE-2023.